

The Evolution of Adaptive LLM-Driven Persuasion: Measuring Behavioural and Societal Consequences

A Research Proposal

Ebrahim Barzegary
ESSEC Business School

Tuck Siong Chung
ESSEC Business School

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Abstract

Marketing is undergoing a fundamental shift from individually crafted messages toward continuous, data-driven optimisation pipelines. Firms increasingly rely on adaptive experiments, A/B testing and black-box methods to refine persuasive strategies, while context-aware algorithms personalise content at the individual level. The emergence of LLM-based optimisation has further accelerated this trend by making it trivial to generate new variants, incorporate them into feedback loops, and update messages in real time.

Despite this rapid technological evolution, we still lack a clear understanding of how such optimisation systems behave over time and what broader consequences they may produce. In particular, it remains unknown how algorithms that learn from human reactions to urgency cues, scarcity framing, FOMO messages, and other persuasive or potentially harmful techniques evolve as they receive more feedback. It is also unclear how these adaptive systems may affect individuals' mental states, behaviour, or well-being—especially when optimisation pressures incentivise the repeated use of techniques that exploit psychological vulnerabilities.

To address these open questions, we propose to build the *Wonderful World of Manipulative Marketing* (WWMM), an interactive research platform in which users engage with marketing “worlds” that employ a range of common persuasive techniques. Content in WWMM will be dynamically updated by an LLM-empowered optimisation algorithm (e.g., LOLA), allowing us to observe how messages shift in response to user reactions and how the system converges—or escalates—over time. By collecting behavioural responses and structured feedback from users, we will examine how different techniques impact individuals, how optimisation dynamics vary across user groups, and whether certain populations experience disproportionately negative outcomes.

This project aims to generate foundational evidence on the societal and psychological impacts of adaptive optimisation in digital marketing. It will provide (i) an experimental environment for auditing optimisation-driven content systems, (ii) empirical insight into the evolution and trade-offs of LLM-assisted persuasive design, and (iii) early detection tools for identifying heterogeneous effects and vulnerable groups. The platform will also serve as an educational resource, helping the public understand the mechanisms of modern optimisation technologies and their potential implications for digital well-being.

1 Introduction and Motivation

Marketing is undergoing a profound transformation. For most of its history, marketing was a human-centered craft: messages were designed by people who understood cultural context, social norms, and the consequences of their decisions. Human-designed persuasion, even when powerful, was interpretable, traceable, and constrained by shared ethical and social expectations.

Today, this landscape has shifted toward large-scale algorithmic systems. Modern marketing relies on opaque optimisation pipelines trained to maximise company-centered objectives such as engagement, click-through rate, or sales. These systems run continuous experiments, update their parameters in real time, and refine persuasive strategies with a speed and granularity that surpass human capabilities. Everyday interventions—from personalised product recommendations to website flows—are shaped by countless automated tests that prioritise short-term business KPIs, while providing little visibility into their broader psychological, societal, or environmental consequences.

A second major shift is the emergence of context-aware and LLM-based optimisation. Traditional persuasive techniques targeted broad human tendencies, such as urgency creation or confirmshaming. While these non-personalised strategies can already exert considerable pressure, modern models personalise content at the individual level. They can infer which emotional patterns, vulnerabilities, or behavioural histories increase responsiveness, and they can adaptively generate new content variants at near-zero marginal cost. For instance, an algorithm may learn that a user searching for “coping with loneliness” is more susceptible to guilt-based or emotionally charged cues. Because these systems operate as black boxes and update themselves continuously, neither firms nor users can easily detect when deeply personal signals enter the optimisation loop. As a result, LLM-driven techniques risk amplifying psychological sensitivities in ways that are both more intrusive and more harmful.

A growing body of evidence links such systems to social and psychological harms, including reduced attention span, heightened loneliness, compulsive usage patterns, negative self-comparisons, and persistent feelings of urgency and stress. Algorithmic optimisation can also reinforce overconsumption cycles and shape behavioural trajectories that undermine long-term autonomy and well-being. As personalisation deepens and optimisation becomes increasingly embedded in daily life, understanding these consequences becomes essential for safeguarding individual rights and societal welfare.

Addressing these challenges requires more than isolated empirical studies. The complexity, adaptivity, and scale of modern optimisation pipelines necessitate a research environment capable of continuous experimentation, long-term observation, and transparent evaluation. To this end, we propose to develop the *Wonderful World of Manipulative Marketing* (WWMM), an interactive research platform designed to study how adaptive, LLM-empowered optimisation systems evolve, how they influence behaviour, and how their impacts vary across individuals and groups. The WWMM ecosystem provides the infrastructure needed for sustained scientific inquiry, educational engagement, and societal impact.

Research Objectives

Our research and development objectives are therefore to:

1. **Characterise the dynamics of adaptive optimisation systems.** Examine how LLM-powered algorithms modify persuasive content in response to user behaviour and how these messages evolve over time.

2. **Evaluate individual-level consequences of algorithmically optimised persuasive techniques.** Measure how exposure to adaptively optimised urgency cues, scarcity framing, confirmshaming, social proof, and other techniques influences behaviour, emotional state, and short-term well-being.
3. **Identify heterogeneous effects and vulnerable subpopulations.** Use tools such as *DivergenceTree* to detect user segments that experience disproportionately negative outcomes, even when average effects appear neutral.
4. **Develop a controlled experimental environment and dataset for auditing optimisation-driven content systems.** Build and maintain the WWMM research platform, including experimental interfaces, data pipelines, and algorithmic evaluation tools that support continuous and reproducible empirical study.
5. **Support education, policy, and public awareness.** Provide an accessible environment where users, students, and policymakers can experience modern manipulative techniques first-hand, understand how adaptive optimisation systems operate, and engage with evidence that informs responsible regulation and digital consumer protection.

Overall, this project seeks to understand not only how adaptive models optimise influence, but also how their behaviour emerges, escalates, and interacts with human psychology, individual vulnerabilities, and societal well-being. By developing a long-term research ecosystem anchored in WWMM, we aim to build a sustainable foundation for evaluating the impact of optimisation technologies and guiding the responsible evolution of digital persuasion in the years ahead.

2 The Wonderful World of Manipulative Marketing Framework

The **Wonderful World of Manipulative Marketing (WWMM)** is an interactive, narrative-based environment designed to reveal how influence works in digital spaces. Instead of communicating marketing concepts through slides or definitions, WWMM uses artistic scenes, playful interactions, and exaggerated versions of real-world techniques. Visitors can explore different worlds, make choices, manipulate elements, and observe how subtle design features steer their behaviour.

This same artistic and immersive environment also serves as a scientific testbed. WWMM allows researchers to modify scenes, vary elements, and introduce different types of persuasive cues. By observing how users interact with these environments, we can study awareness, resistance, emotional response, and behavioural change in a controlled yet realistic setting.

The WWMM website is organised as a set of distinct *worlds*, each one designed around a specific marketing mechanism. For example, the *Deal World* focuses on discounts and pricing techniques, the *Framing World* illustrates how the same information can be presented in gain or loss frames, and the *Urgency World* demonstrates how time pressure and fear of missing out (FOMO) influence decisions. Each world is built as an interactive playground where visitors can explore, click, make choices, and experience how design elements and messages subtly change their behaviour.

A key principle of WWMM is **learning by doing**. In many scenes, visitors are first *tricked* by a manipulative design and only afterwards see an explanation of what happened. For instance, a user might feel pushed to click on a particular button, complete a task more quickly than intended, or accept an offer that looks better than it actually is. After this interaction, the scene reveals the underlying technique and explains how similar mechanisms appear in real digital environments.

This structure makes the lessons concrete and memorable, while also creating clean experimental conditions: each world contains multiple independent components that can be modified and compared.

From a research perspective, this modular design is exceptionally powerful. Because each world consists of independent sections (different texts, buttons, images, layouts), we can run systematic A/B tests and more advanced adaptive experiments. We can vary individual elements (such as the precise wording of a message or the intensity of a visual cue) while keeping the rest of the environment fixed. This allows us to measure how specific manipulations affect engagement, perceived pressure, emotional response, and learning. It also creates an ideal setting for testing adaptive optimisation: the same interface can be used with a fixed baseline design or with an adaptive algorithm that updates content over time.

The *Urgency World* provides a concrete example. Suppose we want to study how FOMO-based messages influence the decision to enter this world. We can present different visitors with different prompts, such as:

- “Don’t miss your chance — the Urgency World is only one click away!”
- “Everyone is already discovering the Urgency World... will you be the last to see it?”
- “Hurry! The Urgency World is moving fast, and you don’t want to fall behind.”
- “If you wait, you may regret not entering the Urgency World when you had the chance.”
- “Don’t be the one left out — step into the Urgency World before it’s too late.”

Some of these prompts use mild FOMO, while others introduce stronger social comparison or loneliness cues. By randomly assigning users to these variants, we can measure both the intended effect (e.g., click-through rate into the Urgency World) and a range of unintended outcomes, such as perceived emotional pressure, stress, or negative feelings after the interaction.

In a second step, LLM-based systems can be used to generate new variants of these messages or to predict which prompts are likely to perform best before deployment. Combined with an adaptive learning algorithm, this creates a closed loop where the model proposes new FOMO prompts, the platform tests them on real users, and the algorithm adapts based on observed behaviour. Within WWMM, this loop can be instrumented not only to maximise engagement, but also to monitor psychological and ethical trade-offs, and to identify subgroups of users for whom a highly optimised FOMO message may become harmful. This makes the platform a unique environment for studying both the power and the risks of personalised, LLM-assisted marketing techniques.

3 Research Questions and Contributions

3.1 Core Research Questions

This project investigates how adaptive, LLM-assisted optimisation systems behave within a controlled environment, how their persuasive strategies evolve over time, and how these dynamics affect individuals and consumer subsegments. We focus on the behavioural, psychological, and societal implications of adaptive optimisation pipelines.

Our core research questions are:

- **RQ1:** How does content generated by an adaptive, LLM-assisted optimisation system evolve over time across different persuasive techniques (e.g., urgency, FOMO, social proof, confirmshaming)? What patterns of escalation, convergence, or novel variants emerge?

- **RQ2:** What trade-offs arise between firm-level optimisation goals (e.g., clicks, engagement, funnel progression) and user-level outcomes (e.g., stress, autonomy, emotional pressure, perceived manipulation, short-term well-being)?
- **RQ3:** How do these effects vary across consumer subsegments? Can divergence-based methods (e.g., *DivergenceTree*) identify groups for whom adaptive optimisation leads to systematically worse outcomes, even when aggregate performance appears improved?

3.2 Expected Scientific Contributions

This project advances scientific understanding of how adaptive, LLM-driven optimisation systems shape human behaviour, interact with psychological vulnerabilities, and generate societal consequences. By observing the real-time evolution of persuasive content, we uncover dynamic patterns that are typically hidden within commercial platforms.

Scientific Contributions.

- **Empirical characterisation of adaptive optimisation dynamics.** We analyse how LLM-assisted optimisation algorithms evolve their persuasive strategies over time, including whether they intensify, diversify, or converge toward specific message patterns in response to user behaviour over iterations.
- **Understanding short-term versus long-term behavioural trade-offs.** We quantify how optimisation targeting immediate engagement influences longer-horizon outcomes such as trust, emotional trajectory, return behaviour, and sustained interaction, clarifying when adaptive algorithms prioritise short-term gains at the expense of user well-being.
- **Firm–consumer trade-offs and heterogeneous effects.** We measure how optimisation focused on business metrics interacts with individual-level outcomes such as stress, emotional pressure, autonomy, and subjective well-being. Using divergence-based methods, we detect user segments that experience disproportionately negative effects despite favourable average outcomes.
- **A unique dataset on algorithmic persuasion and well-being.** The project produces a novel dataset linking behavioural responses, algorithmic decisions, psychological measures, and societal indicators. This resource supports future research on responsible optimisation, digital resilience, and algorithmic fairness.

Applied Contributions.

- **A long-term platform for continuous experimentation and evaluation.** The WWMM ecosystem serves as a persistent research infrastructure enabling repeated experiments, adaptive algorithm audits, and transparent evaluation of emerging optimisation systems.
- **Innovative tools for education and experiential learning.** Through interactive worlds and reflective feedback, WWMM provides a novel pedagogical resource for teaching marketing, behavioural science, digital ethics, and AI responsibility, enabling learning through direct experience rather than abstraction.

- **Evidence for policymaking and regulatory development.** By documenting unintended psychological effects, heterogeneous harms, escalation patterns, and algorithmic dynamics, WWMM offers empirical insights that support emerging regulatory frameworks surrounding persuasive design, algorithmic transparency, and digital consumer protection.
- **Strengthening digital literacy and public understanding.** By revealing to users how they are influenced and explaining the mechanisms behind each technique, WWMM enhances public awareness, helping individuals recognise manipulative design and strengthen autonomy in digital environments.

4 Methodology

4.1 Study Design

Our methodology builds on the modular structure of the WWMM platform, which allows each user to be exposed to different treatments across different parts of the website. This creates a **mixed experimental design**: every visitor experiences multiple independent modules (e.g., urgency world, framing world, deal world), and each module implements its own experimental manipulation.

Participants will be recruited through digital advertisements on platforms such as search engines (e.g. Google) and media platforms (e.g. YouTube) for initial experimentation. Over time, the platform is expected to attract a natural stream of organic visitors. As users navigate the site, they encounter different manipulative or persuasive elements depending on their assigned condition (baseline or adaptive). Throughout the journey, we collect both behavioural measures (e.g., time spent, click patterns, bounce rate) and self-reported responses embedded seamlessly within the experience (e.g., emotional pressure, stress, perceived autonomy). This combination provides a rich dataset capturing both observable behaviour and subjective experience.

Each module includes a **baseline design**, created by humans, and an **adaptive condition**, where the content is updated over time using an LLM-assisted optimisation process. This structure enables us to compare static and adaptive strategies within the same environment and to observe how the adaptive content evolves.

4.2 Metrics

To capture the impact of different content variants and optimisation strategies, we rely on a combination of self-reported and behavioural metrics. The goal is not to limit the project to a fixed set of measures, but to build a flexible framework that can evolve throughout the research period.

Self-Reported Measures. During the interaction, users answer short embedded questions that assess their immediate experience. These measures include perceptions of pressure or manipulation, emotional response, autonomy, and other indicators that are relevant for understanding mental health and individual well-being. These self-reports allow us to observe how different content designs influence subjective states and perceived fairness.

Behavioural Measures. We also record a range of behavioural signals, such as how users navigate the platform, how long they spend on different modules, and their responses to persuasive cues. These measures reflect both short-term outcomes (e.g., clicking on a button) and longer-term

outcomes (e.g., return behaviour, sustained engagement), making it possible to evaluate company-oriented objectives alongside individual-level effects.

Multiple Outcome Dimensions. Our metrics span three perspectives:

- company-oriented outcomes (e.g., engagement, progression),
- individual outcomes related to stress, autonomy, and emotional experience,
- broader societal considerations such as digital trust, perceived autonomy, and perceived manipulation.

Throughout the research period, we will refine, expand, and validate these metrics to identify which measures best capture the behavioural, psychological, and societal implications of algorithmically generated marketing content.

4.3 Analysis Plan

Our analysis plan consists of five components, each designed to capture a different aspect of how algorithmically generated content affects behaviour, well-being, and firm-oriented outcomes. Together, these analyses provide a comprehensive understanding of average effects, heterogeneous impacts, and key trade-offs at both the individual and subgroup level.

1. Average Performance Comparison. We begin by comparing the average performance of the baseline and adaptive conditions across a broad set of outcomes. This includes company-oriented measures, behavioural responses, and self-reported experiences. These comparisons establish baseline differences in how static versus adaptive content shapes user behaviour.

2. Heterogeneous Treatment Effects. Next, we study how the effects of the adaptive condition vary across different subpopulations using algorithms such as Generalized Random Forests (GRF). We identify groups for whom certain content variants or optimisation choices lead to stronger, weaker, or adverse effects. This step allows us to highlight behavioural patterns that would remain hidden when focusing only on average outcomes.

3. Trade-Offs Across Outcomes and Populations. We analyse how the adaptive system balances different objectives, such as short-term behaviours (e.g., clicks or immediate engagement) and long-term outcomes (e.g., time spent, return behaviour, emotional fatigue), as well as firm-oriented metrics versus consumer well-being. Crucially, we evaluate how these trade-offs vary across subpopulations. Some groups may benefit from improved engagement without cost, while others may experience increased stress, reduced autonomy, or negative emotional responses. Using methods such as *DivergenceTree*, we identify segments for whom these trade-offs become particularly unbalanced, revealing where optimisation redistributes value and risk across different types of users.

4. Ethical and Interpretive Assessment. Finally, we contextualise our empirical findings through an ethical lens. We assess whether specific optimisation strategies disproportionately affect vulnerable subgroups, whether certain persuasive techniques raise concerns about manipulation, and where potential harm emerges. This component links the empirical results to broader discussions about responsible marketing, transparency, and digital well-being.

5 Work Plan and Timeline

The project is organised into five sequential phases, each reflecting a major stage of development and research. Together, these phases provide a clear roadmap from initial design to final dissemination.

- **Phase 1: Design of Worlds, Baselines, and Metrics.** Define the detailed structure of the WWMM worlds (e.g., urgency, framing, deals, social proof), create human-designed baseline content, and select initial behavioural and self-report metrics. This phase establishes the foundation for all subsequent experimentation.
- **Phase 2: Platform Implementation and Piloting.** Implement the WWMM platform, integrate logging and embedded survey tools, and conduct pilot tests to validate user flow, data integrity, and the functionality of manipulative elements. Adjustments are made based on pilot outcomes.
- **Phase 3: Experimental Launch.** Begin formal data collection by deploying both baseline and adaptive conditions across multiple worlds. This phase includes recruitment of participants through digital advertisements and the onboarding of organic users. The adaptive algorithm begins updating content based on user interactions.
- **Phase 4: Analysis and Trade-Off Evaluation.** Conduct the analyses outlined in the methodology section, including average performance comparisons, heterogeneous treatment effects, trade-off analysis across metrics, and subpopulation detection using divergence-based methods such as *DivergenceTree*.
- **Phase 5: Dissemination and Integration.** Prepare academic papers, policy-oriented summaries, and public-facing materials. Incorporate WWMM into teaching activities and strengthen the platform for long-term use in future research projects.

6 Impact

6.1 Scientific Impact

This project creates a rigorous empirical foundation for understanding how adaptive, LLM-assisted optimisation systems evolve and how their persuasive strategies affect individuals and groups. By observing real-time message evolution within a controlled environment, WWMM makes it possible to study the dynamics of algorithmic persuasion—phenomena that remain largely invisible in commercial platforms and impossible to isolate in static experimental designs.

The resulting dataset links behavioural responses, optimisation trajectories, psychological measures, and societal indicators, enabling systematic analysis of:

- how LLM-based optimisation algorithms evolve persuasive strategies over time;
- which optimisation pathways produce behavioural or emotional trade-offs for users;
- how effects vary across subpopulations, revealing heterogeneous impacts and potential vulnerabilities;
- the interaction between firm-oriented optimisation and user well-being metrics at short and long time horizons.

This advances research in marketing science, behavioural science, information systems, algorithmic fairness, and AI ethics. More broadly, it introduces a new experimental paradigm for studying optimisation-driven digital environments and their influence on human behaviour.

6.2 Societal Impact

WWMM strengthens public understanding of how modern optimisation systems operate and how they shape digital experiences. By allowing users to interact directly with persuasive techniques—and by transparently revealing the mechanisms behind them—the platform enhances digital literacy and fosters critical awareness of algorithmic influence.

This experiential transparency encourages healthier digital habits, empowers users to recognise potentially harmful designs, and supports greater autonomy in online environments. Furthermore, by documenting how adaptive, personalised optimisation can amplify stress, emotional pressure, loss of autonomy, or negative behavioural trajectories, the project contributes to broader societal conversations about digital well-being and responsible technology deployment.

WWMM thus helps shift public expectations toward more ethical design practices and increases societal demand for responsible, human-centered optimisation.

6.3 Policy Impact

The project provides timely, actionable evidence for policymakers working on algorithmic transparency, persuasive design regulation, dark pattern mitigation, and digital consumer protection. Our results clarify:

- how adaptive optimisation systems influence behaviour, emotions, and societal outcomes;
- which optimisation strategies create harmful trade-offs or escalate persuasive intensity;
- which user groups experience disproportionate negative effects under fully personalised persuasive systems.

The WWMM platform itself functions as a controlled testbed where regulators and researchers can experiment with risk scenarios, evaluate algorithmic decisions, and observe how adaptive optimisation unfolds in real time. This evidence can support emerging regulatory frameworks related to responsible AI, ethical marketing, and the boundaries of acceptable digital influence, helping policymakers craft guidelines grounded in behavioural and empirical insight.

7 Budget and Justification

The proposed budget covers the essential components required to develop, operate, and experimentally evaluate the WWMM platform. Costs are kept intentionally lean, as a significant portion of the technical and research work will be carried out internally, with substantial contributions from volunteer developers, motivated students, and research assistants. These contributions help minimise initial development expenses. Nevertheless, building a reliable, secure platform—and maintaining it throughout experimentation—requires core financial support.

The costs outlined here represent the *initial investment* needed to launch and stabilise the WWMM ecosystem. Once the platform is operational, we plan to pursue additional funding opportunities (e.g., internal grants, national programmes, EU calls) to support continuous improvement, long-term expansion, and broader societal engagement.

7.1 Budget Categories and Estimated Costs

- **AI Development Tools (€3,000–€5,000).** Subscriptions and usage credits for essential development tools, including Cursor, Figma, image/video generation platforms, and LLM API usage required for real-time adaptive content generation within the WWMM worlds.
- **Server and Compute Infrastructure for Two Years (€4,000–€6,000).** Hosting costs for running the WWMM platform, serving interactive worlds, executing optimisation algorithms, and maintaining secure databases for behavioural and self-report measures.
- **Advertising and Participant Acquisition (€2,000–€4,000).** Targeted advertisements (e.g., YouTube, Meta Ads) to obtain sufficient traffic for statistically meaningful experiments. Based on cost-per-view estimates, this will generate approximately 150,000–200,000 visitors.
- **Student Assistants and Research Support (€6,000–€10,000).** Compensation for student developers, research assistants, and data analysts who support platform implementation, experiment configuration, and data processing. No external consulting firms will be engaged in the development process.

7.2 Total Estimated Budget

€15,000 – €25,000

This budget reflects a cost-efficient approach that leverages internal expertise, volunteer contributions, and student involvement while ensuring that the platform remains stable, secure, and scientifically robust. The investment supports not only the immediate goals of this proposal but also the longer-term development of the *Marketing for Betterment* ecosystem. As the platform matures, we will seek additional funding to expand its capabilities, incorporate new experimental modules, and maintain continuous improvement.

8 Ethics and Data Protection

The WWMM platform is intentionally designed to expose users to real manipulative techniques in order to study their effects and to build awareness. Because of this intentional design and the values of the Marketing for Betterment Initiative, we commit to ensuring that every manipulative element used for research is deployed in a controlled, transparent, and safe manner that respects user autonomy and emotional well-being.

Transparency and Informed Participation

Users will be clearly informed that they are interacting with a research platform that studies persuasive and manipulative marketing techniques. The consent process will communicate, in simple language, that:

- users may encounter persuasive or emotionally charged design elements,
- some content may be generated or adapted by algorithms,
- their behaviour and responses will be used for research purposes only.

At all times, users may leave the platform, skip questions, or withdraw participation without penalty. After completing the experience, users will receive a concise debriefing explaining:

- the techniques they encountered,
- why those techniques were used in the study,
- how such techniques appear in real digital environments,
- how to protect themselves from similar manipulations online.

This debriefing is central to the educational mission of the Marketing for Betterment Initiative.

Content Safety and Algorithmic Oversight

Although the platform studies manipulative design, we impose strict boundaries on acceptable content. In particular:

- **All algorithmically generated content (including LLM outputs) will be manually reviewed** by the research team before it can be served to participants. No generated content will be directly shown to users without human oversight.
- **No content will target sensitive personal characteristics, trauma-related cues, or vulnerable populations.** For example, the system will never generate messages referencing loneliness, mental health struggles, financial distress, or personal crises—even if such cues could increase engagement.
- Adaptive optimisation algorithms will **only** operate on a safe, curated pool of pre-approved variants. The algorithms may choose among these variants, but they cannot create or deploy new manipulative elements without human approval.
- We will ensure that any manipulative techniques used in experiments remain within mild to moderate intensity, reflecting what users might ordinarily encounter in commercial platforms, without crossing into harmful or distressing territory.
- After completing an interaction, users receive educational feedback that restores autonomy by explaining how and why the manipulation worked.

These safeguards ensure that the platform mimics the structure of manipulative marketing without exposing users to inappropriate or harmful content.

Data Protection and Privacy

All collected data—including behavioural logs and self-reported measures—will be anonymised and linked only to randomised identifiers. No personal identifying information will be collected beyond what is necessary to operate the platform.

Data will be stored securely on encrypted servers for the duration of the project and handled in accordance with GDPR and institutional guidelines. The database will not include sensitive personal attributes, and no attempts will be made to infer such information from user behaviour or responses.

Broader Responsibility

As part of the Marketing for Betterment Initiative, this project aims not only to study manipulation but also to reduce its negative impact in society. Ethical safeguards are therefore central to the platform’s design. By combining controlled exposure, transparent explanation, and user empowerment, WWMM fosters both scientific insight and public understanding while maintaining high ethical standards.

9 Future Directions

The WWMM platform is designed as a long-term component of the broader *Marketing for Betterment* Initiative, whose mission is to steer marketing toward societal and individual well-being, and environmental responsibility. Within this initiative, WWMM serves two essential roles: it acts as an awareness-creation tool that helps individuals understand how digital influence works, and it functions as a scientific research platform that enables rigorous investigation into the effects of optimisation algorithms.

Looking ahead, the platform can be extended and strengthened in several ways:

- **Expanding marketing mechanisms and environments.** Additional worlds can be created to cover new marketing techniques, emotional triggers, and interface designs, increasing the breadth of behaviours that can be studied and taught.
- **Connecting WWMM to real-world digital environments.** Future work may integrate WWMM with browser extensions or other interfaces, allowing researchers to study how insights transfer from controlled environments to real online contexts.
- **Developing resilience-building interventions.** The platform can be used not only to reveal manipulative techniques but also to test interventions designed to help users resist harmful persuasive strategies and make more informed decisions.
- **Supporting broad research and societal collaboration.** WWMM is positioned to serve as a shared research infrastructure for collaborations with external academics, regulators, NGOs, and digital rights organisations working on transparency, fairness, and digital well-being.

By expanding both its educational and scientific functions, WWMM can contribute to a future in which manipulative marketing techniques become less effective, transparent alternatives become more common, and firms face public demand for responsible and ethical use of optimisation technologies. In this way, the project lays the foundation for a sustained research and societal agenda centred on aligning digital marketing with human well-being.